

Caribbean Coral Reef Institute (CCRI) Water Quality Monitoring Lab University of Puerto Rico – Mayagüez Campus Department of Marine Sciences			
Document Title	Document Number	Version	
SOP Chlorophyll-<i>a</i>	WQP 010	1.1	

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REVISION HISTORY			
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1.0	5/10/23	Initial Release	TAC
1.1	8/28/23	Switched from filtration method to centrifuge method to improve results and reduce materials. Corrected minor grammatical errors. Added 0.1 µg/L limit of detection.	TAC

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1. Title

Standard Operating Procedure for measuring Chlorophyll-*a* in seawater

2. Applicability

This SOP applies to the determination of Chlorophyll-*a* in seawater from shallow (<30 m) coral reef sites around Puerto Rico.

3. Purpose

- 3.1 Provide laboratory-specific instructions for the CCRI Water Quality Monitoring Lab personnel to analyze Chlorophyll-*a* in seawater samples from shallow coral reef sites around Puerto Rico.
- 3.2 Provide water quality data to be uploaded to the EPA WQX database for use by state and federal agencies and any other parties interested in coral reef seawater Chlorophyll-*a* data following the Quality Project Plan.

4. Method

- 4.1 Standard Method 10200 H (Baird et al., 2017).

5. Health and Safety

- 5.1 Acetone is considered a chemical hazard that may be toxic to biological systems if ingested or inhaled. The grinding of filters should be performed in a fume hood due to the volatilization of acetone.
- 5.2 Special handling procedures are listed in the Material Safety Data Sheet (MSDS).
- 5.3 Hydrochloric acid is also a chemical hazard and care should be taken when preparing the solution. The reactivity of HCl with water or with tissue through contact may cause injury. Acid solutions should be prepared in a fume hood to avoid inhalation. Special handling procedures are also listed in the Material Safety Data Sheet (MSDS).

6. General Precautions

- 6.1 Avoid contact with light, which causes degradation of chlorophyll-*a*.
- 6.2 Keep samples on ice or at 4°C and in the dark for no longer than 24 hours before filtering.

7. Training

- 7.1 Personnel shall receive training on how to complete the corresponding analysis steps and forms for this SOP.

8. Required Materials

- 8.1 Consumables
 - 8.1.1 Aluminum Foil, Standard-Gauge Roll
 - 8.1.2 BD Syringe with Luer-Lok Tips (20 mL)
 - 8.1.3 Elite Adjustable-Volume Pipettes

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- 8.1.4 Fine Precision Medium Tipped Tweezers/Forceps
- 8.1.5 Finntip Filter 5mL, sterile
- 8.1.6 1 L Amber Glass Bottles (for sampling)
- 8.1.7 Fluorepolymer resin policeman
- 8.1.8 Glass Cuvette (10 mm path length) (x2)
- 8.1.9 Glass Vacuum Filtration Device
- 8.1.10 Graduated 15mL Centrifuge Tubes with caps (x2)
- 8.1.11 VWR Centrifuge with 15 ml rotor
- 8.1.12 Rack for holding the 15 ml tubes
- 8.1.13 Wheaton Overhead Stirrer
- 8.1.14 PYREX Potter-Elvehjem Tissue Grinder with PTFE Pestle
- 8.1.15 Vacuum Filtration Pump
- 8.1.16 Whatman GF/F Glass Microfiber Filters
- 8.1.17 Millipore Ultrapure Water
- 8.1.18 Orion Aquamate 8100 UV-Vis Spectrophotometer

8.2 List of required chemicals

Compound	CAS No.	Safety classification	
Hydrochloric acid, 0.1N, HCl	7647-01-1	GHS05, GHS07	Warning/Danger – Corrosive, Irritant
Acetone	67-64-1	GHS05, GHS07, GHS09	Warning/Danger – Corrosive, Irritant, Environmentally Damaging
Magnesium carbonate (MgCO ₃)	23389-33-5	---	Not a hazardous substance or mixture
Chlorophyll- <i>a</i> standard	479-61-8	GHS05, GHS07, GHS09	Warning/Danger – Corrosive, Irritant, Environmentally Damaging

8.2.4 Saturated magnesium carbonate solution

Reagent	Amount	Instructions
Magnesium carbonate (MgCO ₃)	1.0 g	1. Weigh 1.0-gram magnesium carbonate. 2. Transfer into a 100 mL volumetric flask. 3. Bring to volume with deionized water. 4. Store indefinitely in a 100 mL glass bottle at room temperature.
DI water	100 mL	

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8.2.3 Aqueous acetone solution (90%)

Reagent	Amount	Instructions
Saturated magnesium carbonate solution	100 mL	1. Measure 100 mL of saturated magnesium carbonate solution into a 1000 mL graduated cylinder. 2. Add 900 mL of HPLC grade acetone into a 1000 mL graduated cylinder to reach 1000 mL total volume. 3. Transfer solution into 1 L glass amber storage bottle. 4. Mix well. 5. Store indefinitely in 1L glass bottle at room temperature.
HPLC grade acetone	900 mL	

9. Procedure

9.1 Sample Collection

- 9.1.1 Prepare a Van Dorn bottle to be deployed.
- 9.1.2 Lower the Van Dorn bottle in a slow and steady mode until it gets close to the bottom with caution not to touch the bottom and stir sediment up.
- 9.1.3 Drop the bottle messenger to close the Van Dorn at the desired depth.
- 9.1.4 Pull the Van Dorn bottle back up into the boat.
- 9.1.5 Connect the Tygon tubing to the Van Dorn bottle spigot to transfer water.
- 9.1.6 Push open the Van Dorn bottle's spigot valve to allow some water to flow out before proceeding.
- 9.1.7 Rinse the 1 L amber glass bottle for sample collection three times with water from the Van Dorn bottle.
- 9.1.8 Collect the sample until the bottle overflows.
- 9.1.9 Place the sample bottle inside the cooler on ice (~4°C) until analysis in the lab.
 - 9.1.9.1 The holding time is 24 hours, the samples should be kept in the refrigerator until processing.

9.2 Sample Filtration

- 9.2.1 Remove sample bottles one at a time from the refrigerator to maintain preservation temperatures.
- 9.2.2 Place a 47 mm glass fiber filter on the filter funnel.
- 9.2.3 Add 2 mL MgCO₃ solution to the sample seawater immediately before filtration.
- 9.2.4 Filter 1.0 L of sample applying vacuum until all sample water is processed.
 - 9.2.4.1 Turn off the vacuum after the sample is processed.
- 9.2.5 Record the total volume of sample filtered for the given sample ID in the chlorophyll-*a* sample notebook.
 - 9.2.5.1 Pour filtrate into a 1000 mL graduated cylinder to record volume.

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9.2.5.2 If the volume exceeds 1000 mL, empty the cylinder and measure the remaining filtrate volume with the cylinder.

9.2.5.3 Rinse the sample bottle with ~20 mL of Millipore Ultrapure water

9.2.6 Turn on the vacuum, and filter the rinse water through the filter.

9.2.6.1 Rinse the graduated cylinder with ~20 mL of Millipore Ultrapure water

9.2.7 Turn off the vacuum and carefully remove the filter tower.

9.2.8 Carefully remove the filter using tweezers.

9.2.9 Fold the filter in half.

9.2.10 Fold the filter in half again.

9.2.11 Place the folded filter into an aluminum weighing dish.

9.2.12 Cover the filter with aluminum foil.

9.2.13 The prepared filter in aluminum foil may be stored in a labeled, sealed plastic bag in the freezer at -20°C for up to 28 days before sample pigment extraction (Section 9.3).

9.3 Sample Pigment Extraction

9.3.1 Take the filter with a tweezer and place it into a grinding tube.

9.3.1.1 While working inside the hood, add 2-3 mL of 90% aqueous acetone solution.

9.3.1.2 Grind for 1 minute at 500 rpm (number 4 in the speed control).

9.3.2 Transfer the extract to a 15 mL centrifuge tube.

9.3.2.1 Use the fluoropolymer resin policeman to drag the contents that were shredded into the tube.

9.3.2.2 Rinse all filter residue into the centrifuge tube with a few ml of aqueous acetone (90%) using a pipette of 1 mL.

9.3.2.3 Bring the final volume up to 10 mL.

9.3.2.4 Record the extract volume in the lab notebook.

9.3.2.4.1 The 47 mm filter takes up 0.1 mL and should be subtracted from the total volume in the pipette.

9.3.3 Cap the centrifuge tube, label, and place in the dark at 4°C for at least 2 hours.

9.3.4 Remove the tubes from the refrigerator.

9.3.5 Place the tubes into the centrifuge and ensure that the rotor is balanced with equal tube weight directly opposite each tube.

9.3.6 Run the centrifuge for 20 minutes at 500 G (3000 rpm).

9.3.7 Decant the extract into a clean 15 mL tube.

9.3.8 Measure and record the extract volume in the lab notebook.

9.3.9 Use 3.0 mL of each extract for spectrophotometric analysis (Section 9.4)

9.4 Spectrophotometric determination of chlorophyll-*a*

9.4.1 Turn on the Orion AquaMate 8100 UV-Vis spectrophotometer.

9.4.2 Ensure that the blank, standard, and samples are at room temperature.

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- 9.4.3 Use the touch screen to swipe to the third page and select “Multi-Wavelength.”
- 9.4.4 Select “Chlorophyll-a.”
- 9.4.5 Verify that the wavelength table reads 664.0, 665.0, and 750.0 nm.
- 9.4.6 Select “Continue.”
- 9.4.7 Add 3 mL of 90% aqueous acetone solution at room temperature to a 1 cm cuvette and cover with a cap.
- 9.4.8 Open the lid, place the cuvette into the sample holder, and close the lid.
- 9.4.9 Select “Blank” to analyze the acetone solution as the sample blank.
- 9.4.10 Remove the cuvette from the spectrophotometer.
- Standard:
 - 9.4.11 Select the “Sample” text, replace it with the Turner Designs Chlorophyll-*a* standard ID, and select “Done” on the keyboard.
 - 9.4.12 Carefully break the Turner Designs Chlorophyll-*a* standard ampule in subdued light and transfer the contents to 2 x 10 mL tubes.
 - 9.4.12.1 The standard may be stored for up to 1 month at -20°C.
 - 9.4.13 Transfer 3 mL of the chlorophyll-*a* standard into a clean 1 cm cuvette using a volumetric pipette and cover with a cap.
 - 9.4.14 Open the lid, place the cuvette into the sample holder, and close the lid.
 - 9.4.15 Select “Measure” to analyze the absorbance of the standard.
 - 9.4.16 Record the absorbances at 664 nm, 665 nm, and 750 nm for the standard in the chlorophyll-*a* sample notebook.
 - 9.4.17 Remove the cuvette from the spectrophotometer and discard acidified standard extract.
 - 9.4.18 Add 0.1 mL (100 µl) of 0.1N HCl with a 100 µl graduated pipet to the cuvette.
 - 9.4.19 Cover the cuvette with the cap and gently invert to mix.
 - 9.4.20 After 90 seconds, return the cuvette to the spectrophotometer.
 - 9.4.21 Select “Measure” to analyze the absorbance of the acidified standard.
 - 9.4.22 Record the absorbances at 664 nm, 665 nm, and 750 nm for the acidified standard in the chlorophyll-*a* sample notebook.
 - 9.4.23 Remove the cuvette from the spectrophotometer.
 - 9.4.24 Calculate Chlorophyll-a concentrations (Section 9.5) for the standard to ensure that the concentrations match the reported values ($\pm 20\%$ or ± 0.2 mg/L) before proceeding to Section 9.2.4 for sample analyses.
- Samples:
 - 9.4.25 Select the “Sample” text, replace it with the Sample Name, and select “Done” on the keyboard.
 - 9.4.26 Transfer 3 mL of the sample clarified extract from Section 9.3 into a 1 cm cuvette using a volumetric pipet (1-10 mL pipette) and cover with a cap.
 - 9.4.27 Open the lid, place the cuvette into the sample holder, and close the lid.

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- 9.4.28 Select “Measure” to analyze the absorbance of the sample clarified extract.
- 9.4.29 Record the sample extract absorbances at 664 nm, 665 nm, and 750 nm for the respective sample ID in the chlorophyll-*a* sample notebook.
- 9.4.30 Remove the cuvette from the spectrophotometer.
- 9.4.31 Add 0.1 mL (100 µl) of 0.1N HCl with a 100 µl graduated pipet to the sample cuvette.
- 9.4.32 Cover the cuvette with the cap and gently invert to mix.
- 9.4.33 After 90 seconds, return the cuvette to the spectrophotometer.
- 9.4.34 Select the “Sample” text, replace it with the Sample Name_a to reflect the acidified measurement, and select “Done” on the keyboard.
- 9.4.35 Select “Measure” to analyze the absorbance of the acidified sample clarified extract.
- 9.4.36 Record the acidified sample extract absorbances at 664 nm, 665 nm, and 750 nm for the respective sample ID in the chlorophyll-*a* sample notebook.
- 9.4.37 Remove the cuvette from the spectrophotometer and discard acidified sample extract.
- 9.4.38 Repeat steps 9.14.24 to 9.4.37 for additional sample extract solutions.
- 9.4.39 Select the “X” in the upper left corner and “end experiment” to end the sample measurements.
- 9.4.39.1 The data are automatically saved with the name, date, and time of analyses on the instrument.
- 9.4.40 Turn off the Orion AquaMate 8100 UV-Vis spectrophotometer.
- 9.5 Determination of Chlorophyll-*a* and Pheophytin-*a* concentrations
- 9.5.1 Use the following formula to calculate chlorophyll *a*:

$$\text{Chlorophyll } \alpha \text{ (}\mu\text{g/L)} = \frac{26.7[(664_b - 750_b) - (665_a - 750_a)] \times V_1}{V_2 \times L} \times 1000$$

where:

26.7 is the absorbance correction

664_b is the absorbance of the sample extract at 664 nm before acidification

750_b is the absorbance of the sample extract at 750 nm before acidification

665_a is the absorbance of the sample extract at 665 nm after acidification

750_a is the absorbance of the sample extract at 750 nm after acidification

V₁ is the volume of the sample extract (mL)

V₂ is the volume of the sample (mL)

L is the width of the cuvette (cm)

1000 converts from mg/L to µg/L

10. Documentation

- 10.1 Field data will be recorded in Field Form (Appendix I).
- 10.2 Calibration data will be recorded in Calibration Form (Appendix F).
- 10.3 Field and Calibration forms shall be kept in the designated file cabinet at CCRI office.

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10.4 Data contained in hard copy forms shall be digitized into the corresponding electronic spreadsheets as soon as possible after the task is completed.

10.5 All spreadsheets and depth profile logs should be uploaded to the internal CCRI Water Quality Database.

10.6 All data will be electronically archived and backed up online.

11. Cleaning and Maintenance

11.1 The grinding of filters during the extraction process should be conducted in a fume hood due to the volatilization of acetone by the tissue grinder.

11.2 Always wear a lab coat, gloves and safety glasses while preparing reagents and performing the method.

11.3 Acetone is a chemical hazard and may be toxic to biological systems if ingested or inhaled. The grinding of filters should be performed in a fume hood due to the volatilization of acetone. Special handling procedures are listed in the Material Safety Data Sheet (MSDS).

11.4 Hydrochloric acid is also a chemical hazard and care should be taken when preparing the solution.

11.5 The reactivity of HCl with water or with tissue through contact may cause injury. Acid solutions should be prepared in a fume hood to avoid inhalation. Special handling procedures are also listed in the Material Safety Data Sheet (MSDS).

11.6 Do not leave samples out of the refrigerator when they are not being worked on.

11.7 Analyze the samples in the order in which they were extracted so that all samples are steeped for approximately the same amount of time.

11.8 The holding time should be between 24 to 48 hours at 4°C, until filtration, and after filtration, 28 days, frozen, until analysis.

12. Troubleshooting

12.1 Acetone is considered a chemical hazard that may be toxic to biological systems if ingested or inhaled. The grinding of filters should be performed in a fume hood due to the volatilization of acetone. Special handling procedures are listed in the Material Safety Data Sheet (MSDS).

12.4 Hydrochloric acid is also a chemical hazard and care should be taken when preparing the solution. The reactivity of HCL with water or with tissue through contact may cause injury. Acid solutions should be prepared in a fume hood to avoid inhalation. Special handling procedures are also listed in the Material Safety Data Sheet (MSDS).

13. QA/QC

13.1 In accordance with the QAPP Table 7, the certified chlorophyll-*a* standard solution should be analyzed each day of analysis to within $\pm 20\%$ or ± 0.2 mg/L.

13.1.1 Limit of detection is 0.1 $\mu\text{g/L}$.

13.2 Any replicate samples should be within $\pm 20\%$ or ± 0.2 mg/L.

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13.3 The Orion AquaMate 8100 UV-Vis Spectrophotometer should be operated and maintained following the manufacturer specifications.

14. References

- 14.1 Baird, R. B., Eaton, A. D., & Rice, E. W. (2017). Standard methods for the examination of water and wastewater (Vol. 23). E. W. Rice (Ed.). Washington, DC: American Public Health Association.
- 14.2 Orion AquaMate 8100 UV-Vis User Manual